"A good hockey player skates to where the puck is. A great hockey player skates to where the puck is going to be." - Wayne Gretzky

Maximizing Profit for Adirondack Growth Capital in a Potential Acquisition of the Peoria Rivermen Minor League Hockey Franchise

OIT 367 - Business Intelligence from Big Data

Final Project

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You may keep our final project.

1. Executive Summary

1.1 Problem Description

We were tasked by Adirondack Growth Capital ("Adirondack" or "AGC") with investigating team revenue streams for the Peoria Rivermen, a minor league hockey team in the Southern Professional Hockey League (SPHL). Adirondack, a private equity firm, is assessing whether to purchase the team, and they desire to understand what levers can be pulled to maximize profit if they make the purchase. The goal of this project is to isolate variables that have the most impact on overall revenues, leading to profit potential. To conduct this analysis, Adirondack provided profit/loss statements for each game played over the last three seasons. These statements include total ticket sales, split between box office sales, team sales, and complementary tickets. Concession revenue is also included. As part of this assignment, our team has created predictive models to estimate future team revenues based on a series of variables including, but not limited to: opponent, day of week, weather, and opposing team record. The goal of the analysis is to assess which variables are significant and to inform a potential acquirer (such as Adirondack) of potential strategies to increase revenues. Based on our modeling, we would then be able to recommend marketing strategies, such as promotions and giveaways, to optimize revenue for games with lower expected revenue.

One of the biggest challenges in this assessment is the relatively limited data set we have to work with -- three seasons of data only equates to 63 home games from which to make validation and training sets. As such, the reliability of our models may not be as robust as projects with far larger numbers of observations to pull from.

1.2 Principal Findings & Recommendation

Using multiple iterations of LASSO (Least Absolute Shrinkage and Selection Operator) models, we found that the principal driver of overall profitability was, unsurprisingly, maximizing ticket sales and concession revenues. Digging further, excluding financial measures within the regressions, we discovered that day of the week, overall team record, and the ratio of complimentary tickets to paid tickets had notable predictive effects. Finally, we looked at the effect that paid and complimentary tickets had on concessions, and discovered that increasing the number of complimentary tickets to paid tickets increases overall concession revenue, in turn likely increasing total revenue. While these results are subject to the limitations of a small data set, initial recommendations include giving away more tickets when overall sales are down to juice concession sales and avoiding Sunday games when possible (subject to scheduling constraints provided by the league).

2. Problem Statement

Adirondack Growth Capital is a private equity firm specializing in investing in companies with between $5 and $50 million in revenue in a variety of industries. The play to acquire the Peoria Rivermen stems from the team being in the sweet spot within their investment portfolio, while also capturing an opportunity to own a team in a sport that the managing director has a passion for -- hockey. Before making the purchase, however, Adirondack seeks to understand the drivers of revenue and profitability for the franchise. One area of concern is that within the past four years, the Rivermen dropped from a relatively more competitive minor hockey league (the American Hockey League, which is a development league for the National Hockey League) to a less competitive one (the Southern Professional Hockey League, which has no NHL affiliations), with a significant drop in revenue. The Adirondack team wants to understand what elements they can optimize to maximize the profit their firm can extract from operating the team, and this understanding will feed into their final bid submitted for the Rivermen.

2.1 Data and Sources

The data we used was provided directly to us by the managing partner of Adirondack Growth Capital, David Kidd. We have included a detailed description of each variable in Exhibit 1.

We were given separate spreadsheets for each of the last 63 home games played by the Rivermen, which we compiled into an master data set. Furthermore, higher level amplifying data was given for seasons prior, but as it was in a different league than the current team, we chose to focus on the more recent information.

We also did independent research to fill out other elements of the data set, namely related to weather and the overall performance of the team. This was compiled from online historical weather data as well as the won-loss record provided on the Rivermen's webpage.

Broadly speaking, each of the individual 63 game event settlements contained the same information. This included ticket sales divided into three categories: revenues from concessions, total attendance, charges for using the facilities and the total revenue due to the Rivermen after all expenses were accounted for. An example is included in Appendix 1.

3. Analysis Methodology

In completing our analysis, we completed the following three tasks, which are described in greater detail below:

1. Data pre-processing**:** aggregated the individual game data into a single dataset.
2. Feature engineering: used the client data to create additional variables for regression.
3. Model development: built and tested several models using LASSO and Logistic Regression techniques

3.1 Data pre-processing

The raw data provided by the Peoria Rivermen primarily consisted of game-level information. The club provided a profit/loss statement for each game, which contains several variables related to attendance, ticket prices, concession revenue, and game information. There are several main challenges in the main data, listed below:

1. Each game's attendance and revenue information was recorded in a separate Excel spreadsheet.
2. Some summary level data was provided for games in each of the past two seasons. However, this data was not complete and required additional data entry to ensure a complete and consistent dataset.
3. There were entries in the data fields that were either blank or "na".
4. Summary level data only included information for Seasons 2 and 3.
5. The data contain both categorical and numerical variables.

In order to ensure a reliable and useable dataset, the following activities were performed on the data:

1. A new Excel spreadsheet was created that listed all games and their associated data in a tabular format.
2. Additional data was manually entered into the pre-supplied summary data to ensure consistency across seasons.
3. Data was aggregated to ensure no blank or "na" fields
4. Summary level data from seasons 2 & 3 were appended to a newly created summarized dataset for season 1. This ensured a full set of data covering all of Peoria's games as part of the SPHL (Southern Professional Hockey League).

5. Categorical variables were binned in R and dummy variables were created for use in regression.
6. We also created additional columns through external research to capture the effects of weather -- both temperature and precipitation -- while also recording the overall games above / below .500
7. We built an interaction variable between complimentary and paid tickets to assess how the ratio impacted our models

3.2 Feature engineering

In order to create the best possible supervised learning model, we created new features to test various models. Since our goal was to find which variables influence revenues the most, we used Balance_to_RM as our response variable.

Many of the potential dependent variables were already present in the raw data. These included attendance, ticket prices, concession revenue, and other game information. As will be shown later in this report, some of these were useful, but many were not.

To further refine our search, we built additional variables that we believe may have predictive power:

- The first was breaking up the different opponents into dummy variables. Specific team rivalries are often known to draw more crowds in professional hockey games, and we wanted to test if this was the case for the Rivermen. As such, OpponentXYZ is the dummy for each opponent played.
- We also split each day of the week into a dummy variable to see if there were any effect due to the weekends or weekdays. Each day of the week, aside from Wednesday, was present in the data. Our dummy variables were entitled DOWeek.
- We also wanted to assess how weather affected overall revenues -- there was a possibility that cold or hot weather, or a large amount of precipitation, would affect attendance and revenues. To analyze this, we looked at historical weather data, creating a MeanTemp variable. We also determined whether 0.5” or more of precipitation fell during a particular home game, and coded this as the “InclementWX” dummy variable.
- To assess the impact of giving away tickets, we also created a complimentary / paid ticket ratio, which resided within the data as “Comp2Paid.”
- Finally, we surmised that the team’s performance would impact attendance and revenue. We went through the win loss ratio for all the

home games, and numerically coded the number of games above or below .500. According to experts such as Harvard Sports, this is one of the most important variables to predict how full will a stadium be.

3.3 Model development

After pre-processing the data and creating new variables to test, we began to build a model. With the limited number of data points (only 63) and many potential dependent variables, we were unable to do a traditional training, validation and test set. Our initial analysis by creating a series of histograms and box plots to determine how the variables interacted with each other. Many were inconclusive, but a few correlations stood out. See Appendix I, Exhibit 3 for these visuals.

To further refine which variables to use, however, we utilized the Lasso technique. This allowed us to do 10-fold cross validations to identify which variables had the greatest impact on overall prediction.

The first Lasso model identified three primary variables that significantly impacted team revenue: the number of paid tickets, the overall receipts, and the total amount from concessions. This seems rather tautological, but it helped drive us to further evaluate the model and determine if there were any other non-monetary levers influencing these underlying numbers. The interesting to observe that price, secondary tickets and comp tickets were left out of these model.

Our next step was to eliminate all the variables that were related to revenues, so we eliminated the following variables: Paid, Secondary, Comp, Total, Drop.Total, AvgPrice, TicketSales, SeasonTickets, ImprovementFee, Receipts, LuxurySuite and Concessions. We did this to access which more indirect measures impacted revenue. Again, we did a Lasso analysis. We found that non-financial measures were significant enough to have an impact. The most significant were whether a game was played on a Saturday or Sunday, what the overall record of the team was, and what the ratio of complimentary tickets to paid tickets were.

However, with a negative sign on the “Comp2Paid” variable from the previous model, we wanted to understand what drove concession sales. That does not necessarily mean that giving more complimentary tickets is the cause of low revenues but, in general, games with bad revenue have more complimentary tickets. Thus, it seemed to us that the negative Comp2Paid variable may have had underlying causes: the team may be

more likely to give more complimentary tickets in a game that it does not expected to sell out.

In the third model, we wanted to see which factors impacted the concession revenues, something that we discovered it was very significant to the overall revenues. Once again, we applied a Lasso methodology to determine what elements impacted concession sales. This time, there were far more relatively significant drivers than the previous two models. Day of the week again mattered, as did record, but some opponents helped to increase concession sales as well. Interestingly, the Comp2pay is positive and significant in this case. This shows that a larger proportion of complimentary tickets help to drive up the concession revenues.

Again, while the small dataset was not sufficient to do a true training, validation and test series, the Lasso model integrates inherent cross validation, giving us some confidence that our models are somewhat predictive.

Once we identified the most significant variables as determined by the Lasso analysis, we regressed the associated independent variables on the response.

4. Results Summary

The final R-code (see appendix 2) was used to build our Lasso and LM regressions. The Lasso graphical findings are in Appendix I, Exhibit 3. For our first Lasso and LM combination, we found that the Response, revenue to the team, was impacted most heavily by Paid tickets, total receipts and Concession sales. However, only the last two elements were statistically significant. The results are below:

Variable	Lasso Coefficient	Significance Level	LM Coefficient	Variable Description
Paid	1.1167	N/A	1.371	Number of Paid Tickets
Receipts	.8771	***	.9513	Total Ticket Revenue
Concessions	.8054	***	1.018	Total Concession Revenue

Following this analysis, we looked to eliminate the financial drivers, and determine what other elements had an impact on revenue. The results are below:

Variable	Lasso Coefficient	Significance Level	LM Coefficient	Variable Description
DOWeekSat	3464.74	***	4898.9	Saturday game
DOWeekSun	-1559.046	***	-5098.1	Sunday Game
Record	179.418	***	469.3	Games above/below .500
Comp2Paid	-173.508	*	-678.5	Ratio of comp to paid tickets

Finally, we assessed what drove concession sales:

Variable	Lasso Coefficient	Significance Level	LM Coefficient	Variable Description
OpponentFlyers	344.15	*	1098.68	Played against the Flyers
OpponentKnoxville	154.85	.	531.96	Played against Knoxville
OpponentMississippi	199.75	*	581.06	Played against Mississippi
DOWeekSun	-658.42	**	-793.92	Sunday game
DOWeekThur	146.47	N/A	415.80	Thursday game
Record	36.65	**	64.19	Games above/below .500
Paid	3.53	***	3.98	Number of paid tickets
Comp2Paid	77.94	***	173.82	Ratio of comp / paid tickets

5. Principal findings, recommendation, and conclusion

5.1 Model interpretation

Using data from the tables in the previous section, the model indicates the following:

- The number of paid tickets, overall concession sales and total receipts drive profitability for the Rivermen. The positive coefficients for all these variables indicate that increasing those values will have a positive impact on profitability.
- Excluding direct financial drivers from the model indicates that the day of the week, the record of the Rivermen, and the proportion of complimentary tickets to paid tickets matters. If a game is played on Saturday, profit will be significantly higher. If a game is played on Sunday, profit will be significantly lower. Furthermore, for each extra game above .500 the Rivermen play, revenue will

increase by about $570. Investing in players to create a good team is a possibility to create higher revenue, as long as the price paid for the player does not outstrip the benefit to the team's record. Finally, the negative coefficient on the complimentary / paid ticket ratio indicates that complimentary tickets are negatively correlated with profitability. As discussed in section 3, this does not mean that complimentary tickets cause lower revenues, because management may give away more tickets if it is know that the game is not going to fill up the stadium.

- Looking at concession sales, a driver of profitability in the opening regression, we find that the most significant drivers are day of the week, record, and attendance metrics. Sundays have significantly lower concession sales. As the team's record gets better, concession sales slightly increase. More paid tickets also equates to slightly higher concession numbers. However, interestingly, as more complimentary tickets are provided in relation to paid tickets, concession sales increase. Perhaps those who do not pay for their tickets are more willing to spend money on ancillary products once inside the gates.

5.2 Prediction

Due to the limited number of data points, we were not able to create a very robust prediction model. However, the Lasso analysis gave us a good sense of which variables drive revenue and concession sales. The scale of each variable may be slightly different than the model analysis, but the direction of movement is likely accurate.

5.3 Recommendations for the Peoria Rivermen

We translate our recommendations to several concrete strategies as the AGC team looks to optimize operations if they do indeed own the Rivermen franchise. Based on our analysis, we recommend strategies such as the follows:

- Negotiate with SPHL owners on scheduling such that all scheduled Saturday games are played in Peoria if possible, so that the Rivermen maximize opportunities to gain ticket and concessions revenue.
- Focus on recruiting strong players to join the Rivermen, preferably at the center position - an aggressive, powerful "enforcer" in center or the wings can increase goal scoring, leading to a winning record (and potentially more ticket revenue, as a substantial percentage of hockey fans come to see fights on ice)
- On nights with historically low concession sales such as Sundays, host "dollar ticket nights" where fans can get seats for a dollar, while helping to drive concession revenue. Another way to package this deal may be to advertise the

game in the Peoria Journal Star for free, by asking fans to bring in a clipping of the advertisement (the Journal Star may give this to the Rivermen for free, as a paid-in-kind arrangement).

- Hire a corporate sales associate to focus on selling box tickets (if available) to small businesses, Scout troops, church groups, etc. at preferential rates, so as to encourage group buying.
- Host promotions on weekday nights, such as pictures with a player, pre-game skating on the ice, slapshot competitions, etc. to drive ticket sales.
- Experiment with hosting concession vendors who would sell more varied fare (e.g., ice cream sandwiches, funnel cake, etc.) on nights where there is a higher percentage of comped tickets, in the hope of capturing more concession revenue.

Appendix I: Exhibits

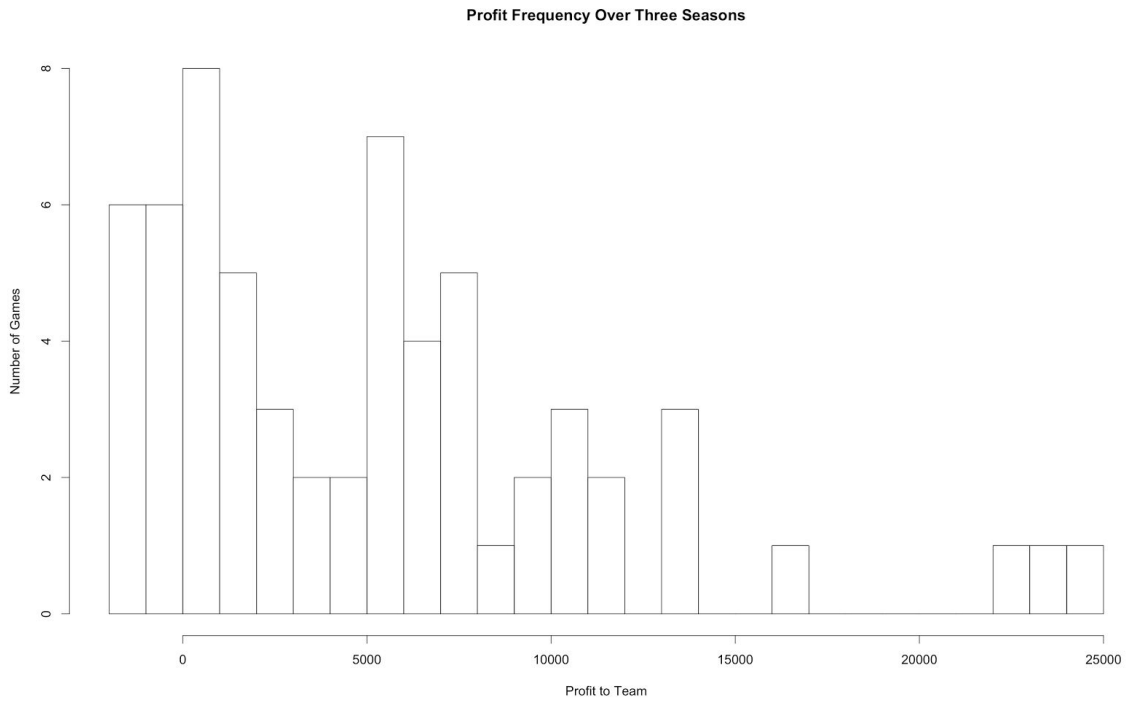
Exhibit 1: Description of data provided by the Peoria Rivermen

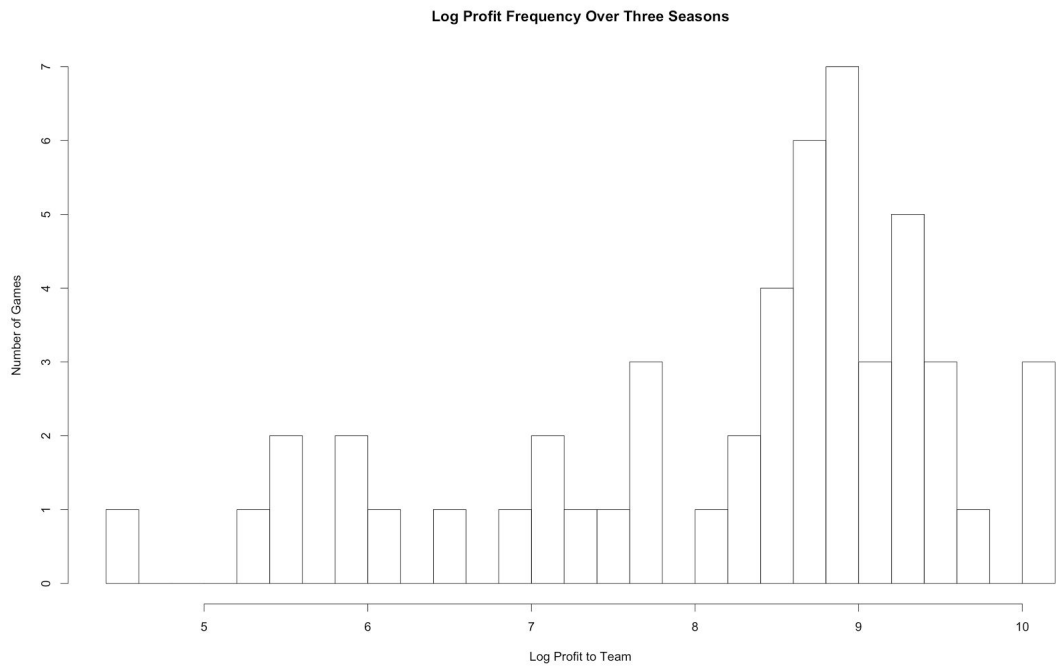
Raw Data	
Variable Name	Variable Description
Opponent	Name of opponent visiting the Rivermen
Season	Season # in the SPHL
Date	Date of the game
Paid	# of tickets purchased at box office
Secondary	# of tickets purchased through the team
Comp	# of complementary tickets given away
Total	# of total tickets distributed
Drop Total	# of actual tickets redeemed / proxy for real attendance
AvgPrice	Average price per ticket
TicketSales	Total ticket sales (in $)
SeasonTickets	Amount deducted from sales for season tickets (season tickets are sold in advance and are backed out of the total to prevent double counting
Concession	Concession revenue share (in $) due to the Rivermen from Peoria Convention Center (PCC).
PCCRent	Rent charged by PCC (in $)
Balance_to_RM (	Total $ due to Rivermen for concessions and tickets

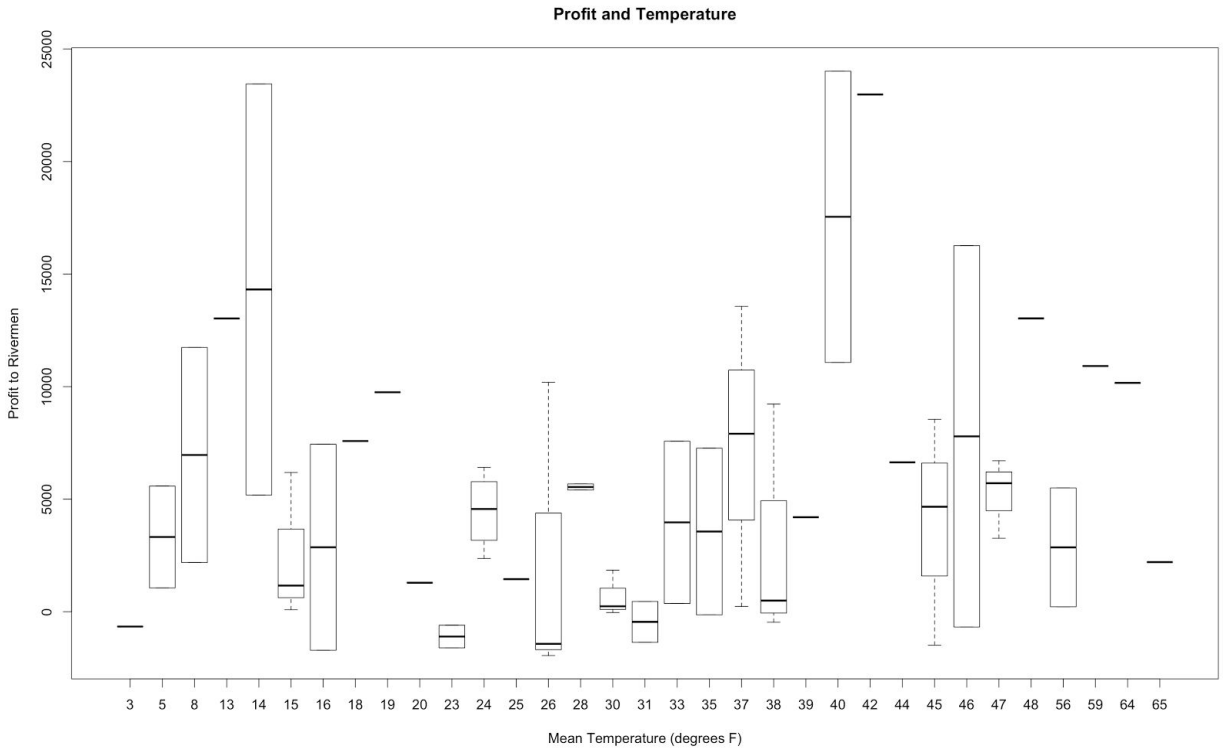
Exhibit 2: Description of aggregated data including additional variables

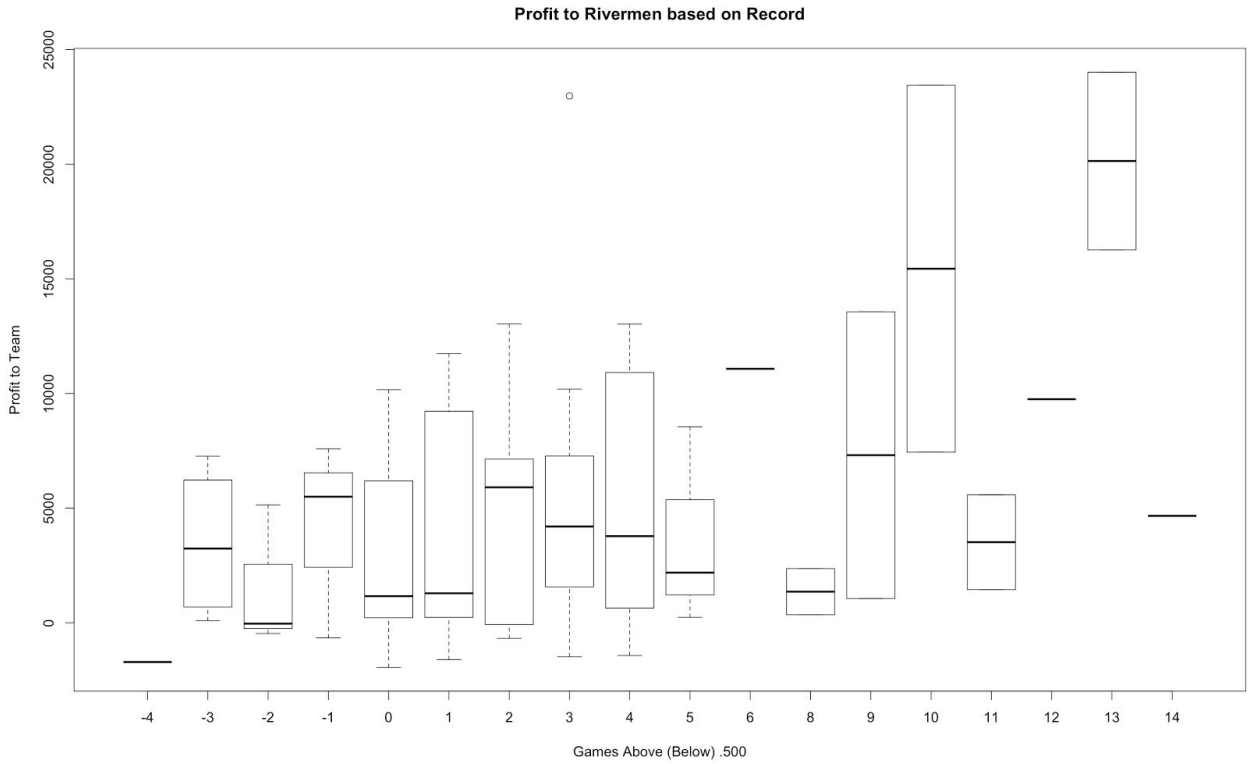
Aggregated Data		
Variable Name	Variable Description	Used?
Opponent	Name of opponent visiting the Rivermen	Yes
OpponentXYZ	Dummy variable for each opponent played, derived from $Opponent	Yes
Record	Number of games above / below (negative) .500	Yes
DOWeek	Day of the week the game is played	Yes
DOWeekSunday - DOWeekSaturday	7 Dummy variables for each day of the week, derived from $DOWeek	Yes
Season	Season # in the SPHL	No
Date	Date of the game	No
InclementWx	Dummy variable indicating >0.5” of precipitation	Yes
MeanTemp	Average temperature on the day of game (in degrees F)	Yes
Paid	# of tickets purchased at box office	Yes
Secondary	# of tickets purchased through the team	Yes
Comp	# of complementary tickets given away	Yes
Total	# of total tickets distributed	Yes
Drop Total	# of actual tickets redeemed / proxy for real attendance	Yes
AvgPrice	Average price per ticket	No
TicketSales	Total ticket sales (in $)	No
SeasonTickets	Amount deducted from sales for season tickets (season tickets are sold in advance and are backed out of the total to prevent double counting	No
Concession	Concession revenue share (in $) due to the Rivermen from Peoria Convention Center (PCC).	Yes
PCCRent	Rent charged by PCC (in $)	No
Comp2Paid	Ratio of Complimentary Tickets to Paid Tickets	Yes
Balance_to_RM (RESPONSE)	Total $ due to Rivermen for concessions and tickets	Yes

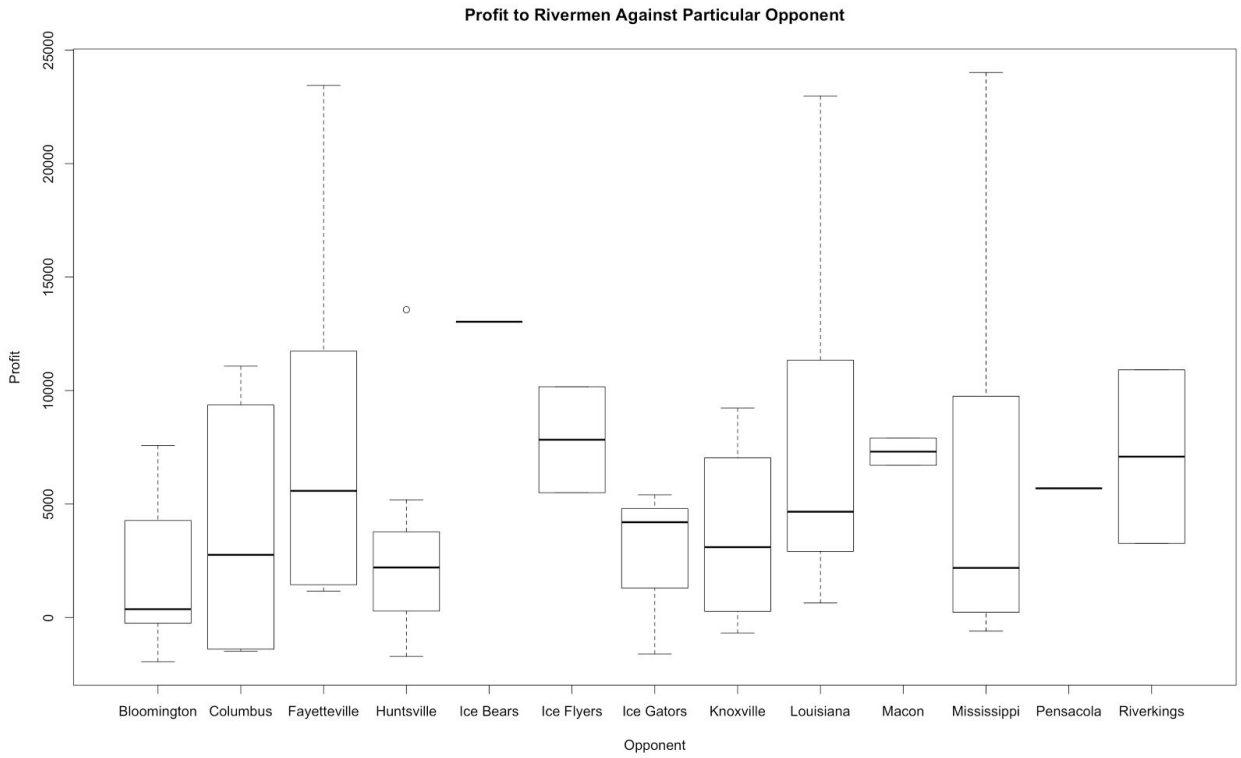
Exhibit 3: Various Histograms and Box Plots

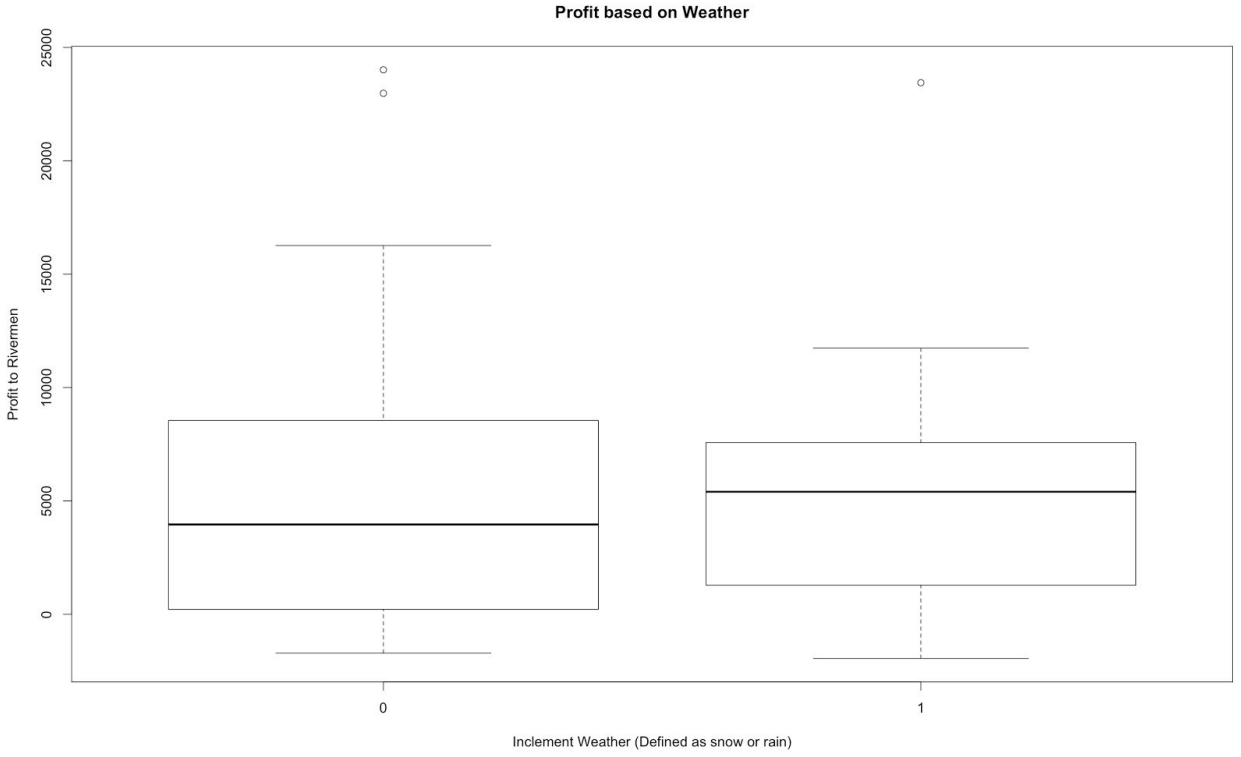





Profit and Temperature
Profit to Rivermen
25000
20000
15000
10000
5000
0
3 5 8 13 14 15 16 18 19 20 23 24 25 26 28 30 31 33 35 37 38 39 40 42 44 45 46 47 48 56 59 64 65
Mean Temperature (degrees F)


Profit to Rivermen based on Record
Profit to Team
25000
20000
15000
10000
5000
0
-4 -3 -2 -1 0 1 2 3 4 5 6 8 9 10 11 12 13 14
Games Above (Below) .500


Profit to Rivermen Against Particular Opponent
Profit
25000
20000
15000
10000
5000
0
Bloomington
Columbus
Fayetteville
Huntsville
Ice Bears
Ice Flyers
Ice Gators
Knoxville
Louisiana
Macon
Mississippi
Pensacola
Riverkings
Opponent


Profit based on Weather
Profit to Rivermen
25000
20000
15000
10000
5000
0
0
1
Inclement Weather (Defined as snow or rain)

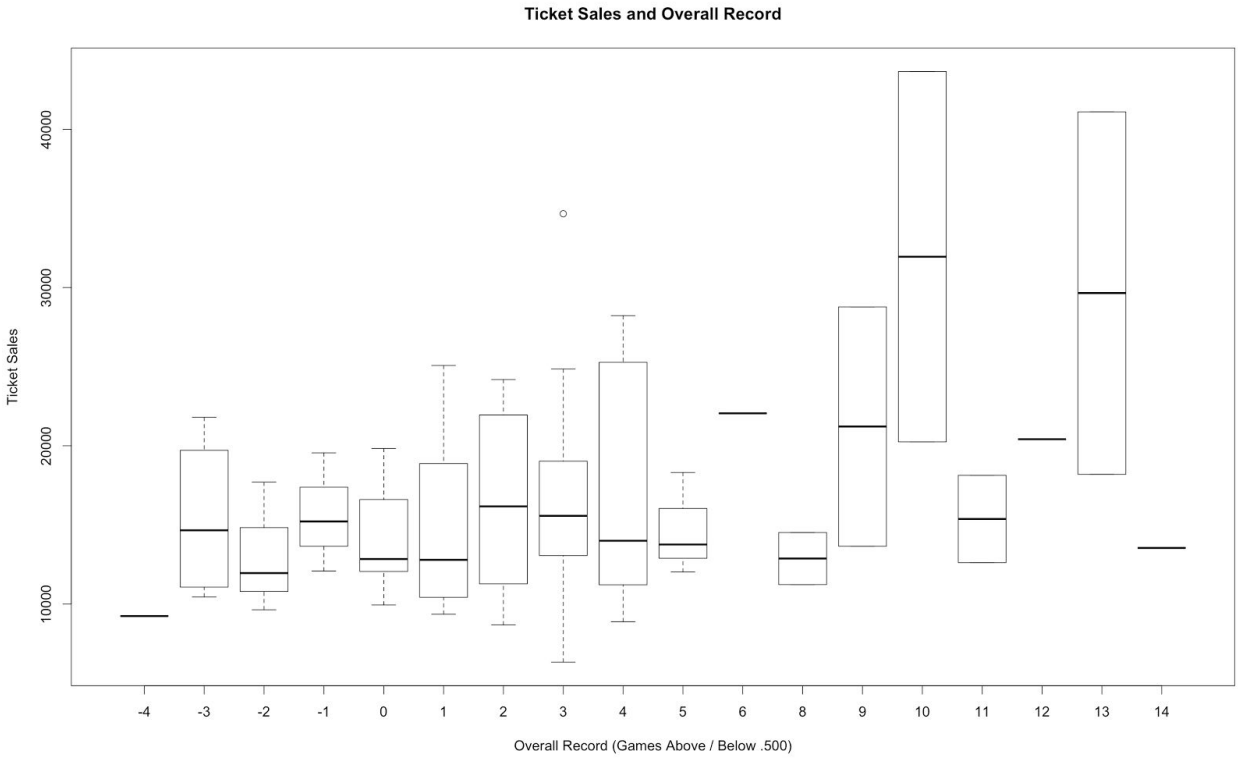

Ticket Sales and Overall Record
Ticket Sales
10000
20000
30000
40000
-4
-3
-2
-1
0
1
2
3
4
5
6
8
9
10
11
12
13
14
Overall Record (Games Above / Below .500)

Exhibit 4: Sample Game Data

Peoria Civic Center
Event Settlement

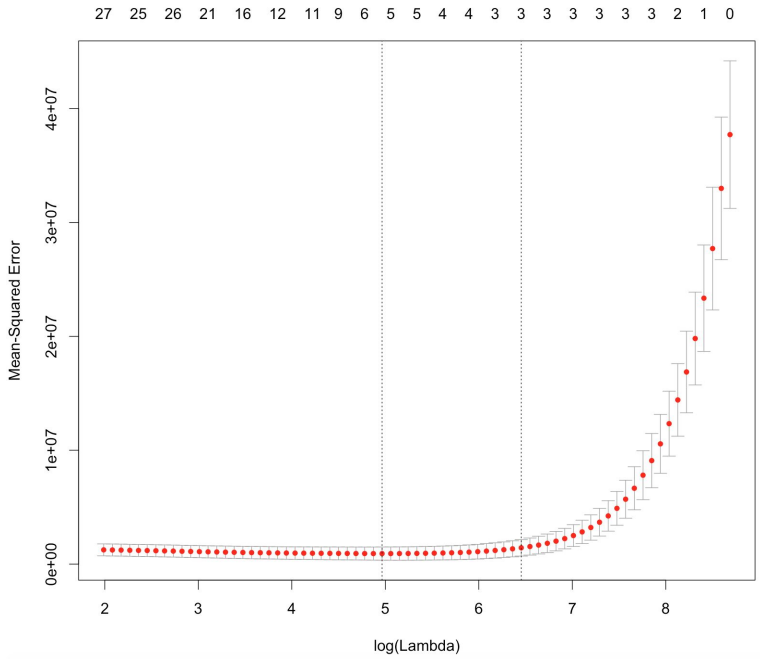
Event Name	**Rivermen vs. Fayettevile**	Event Number	15-175
Date	February 28, 2015	Event Type	Rivermen
Promoter	Rivermen	Cost Center	721

Gross Ticket Sales	$ 48,333.91
Less $1.00 Improvement Fee	$ (3,819.00)
Taxable Ticket Sales	$ 44,514.91
Less Ticket Tax	$ 872.84
Gross After Tax Ticket Sales	$ 43,642.07
Less Season Tickets & Coupons	$ (21,726.61)
Less Secondary Improvement Fee	$ (1,881.00)
Total Receipts	$ 20,034.46
CASH DISTRIBUTION	
Tickets Receipts Less Tax	$ 20,034.46
Due Rivermen	$ 20,034.46
Less Cash Draw	
Net Due Rivermen	$ 20,034.46
Ticket Receipts Due Rivermen	$ 20,034.46
Plus Luxury Suite Tickets	$ 270.00
Plus Facility Fees for first 400 tickets sold	$ 400.00
Total Due Rivermen	$ 20,704.46
Amount Due Rivermen(PCC)	$ 4,855.36
Balance Due Rivermen(PCC)	$ 25,559.82
Less Invoice E00546	$ (122.00)
Balance Due Rivermen(PCC)	$ 25,437.82

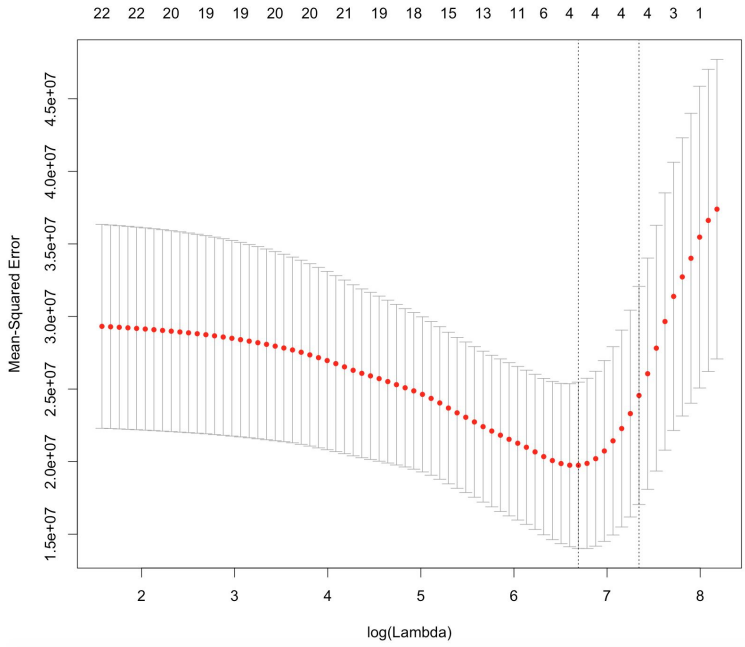
Attendance			Drop	
Paid	1,938		**5,115**	
Paid wo Imp Fee	16			
Secondary	1,881			
Comp	2,802			
Total	6,637			
Ticket Prices	$ 23.00	$ 20.00	$ 18.00	$ 13.00
Average Ticket Price				$ 11.61
Building Fees & Charges				
Rent				$ 7,000.00
(Less labor reductions)				$ (1,065.00)
Bus Parking Passes		2	$ 14.00	$ 28.00
Total Fees & Charges				$ 5,963.00
Concessions Profit Split-Rivermen				$ 8,703.36
Game Rebate				$ 2,115.00
Less PCC Expenses				$ (5,963.00)
Due Rivermen(PCC)				$ 4,855.36
Balance Due Rivermen(PCC) from Prior Game				
This Game				$ 4,855.36
Balance Due				$ 4,855.36

Exhibit 5 - Lasso Results

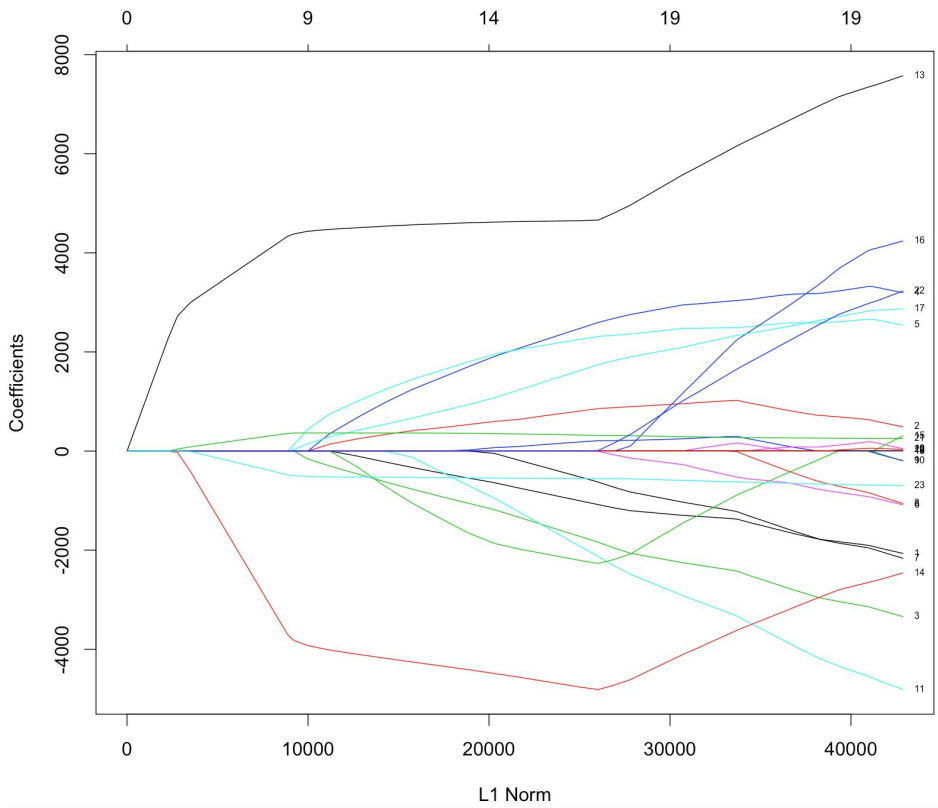
Principle Lasso - Revenue to Team (RESPONSE) and associated Lambda



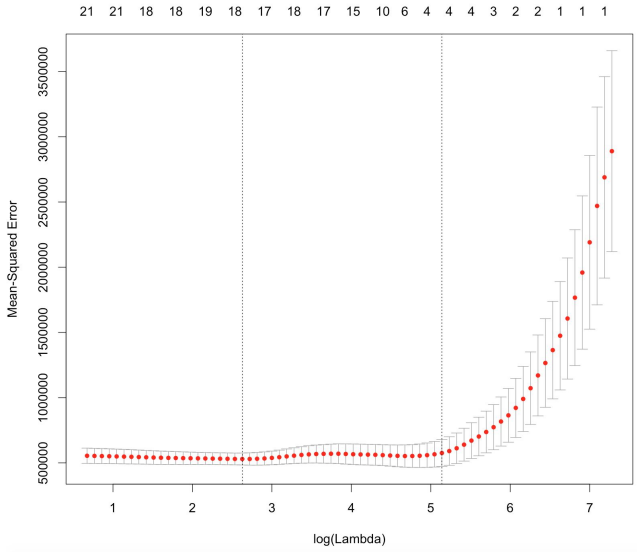
Second Lasso - non-financial drivers of Response



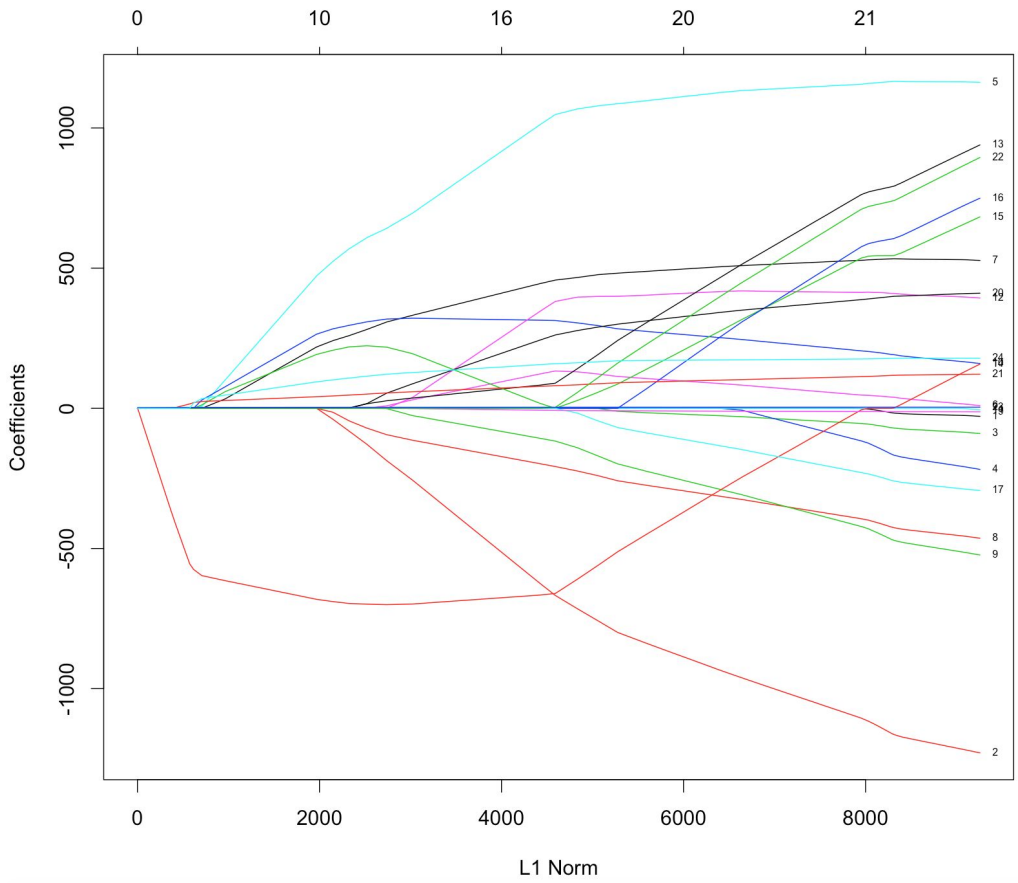
Variable Selection from Second Lasso



Third Lasso - Impact of Independent variables on Concession Sales



Variable Selection on Third Lasso



Appendix II: R-Code

```
rm(list=ls())

source("http://www.stanford.edu/~bayati/oit367/T367_utilities_11.R")
set.seed(1)
#Read the files

#Read the files
setwd("~/Downloads")
data = read.csv("SPHLWx.csv")

data$Comp2Paid = data$Comp/data$Paid

#Bin data
names.categorical.predictors = c("Opponent","DOWeek","Season")
```

```
names.numerical.predictors = colnames(data)[!(colnames(data) %in%
c(names.categorical.predictors,"Balance_to_RM"))]
RESPONSE = as.numeric(data$Balance_to_RM)

numerical.predictors = data[,names.numerical.predictors]
categorical.predictors = data[,names.categorical.predictors]

categorical.predictors.binned = bin.cat.data(data[, names.categorical.predictors])
data.expanded = as.data.frame(cbind(categorical.predictors.binned,
numerical.predictors,RESPONSE))
library(data.table)
setnames(data.expanded,old=c("OpponentIce Bears","OpponentIce Flyers","OpponentIce
Gators"), new=c("icebears","flyers","gators"))
#Randomize the data
data.expanded = data.expanded[sample(nrow(data)),]

data.expanded$Date = NULL
data.expanded$TotalDue = NULL
data.expanded$PCCRent = NULL
#histograms
hist(data.expanded$RESPONSE, breaks=20, main="Profit Frequency Over Three
Seasons",xlab="Profit to Team", ylab="Number of Games")
hist(log(data.expanded$RESPONSE), breaks = 20, main="Log Profit Frequency Over Three
Seasons",xlab="Log Profit to Team", ylab="Number of Games")
hist(data.expanded$Comp)

#boxplots
boxplot(RESPONSE~Opponent,data=data, xlab="Opponent", ylab="Profit",main="Profit to
Rivermen Against Particular Opponent")
boxplot(RESPONSE~Record,data=data.expanded, xlab="Games Above (Below) .500",
ylab="Profit to Team", main="Profit to Rivermen based on Record")
boxplot(RESPONSE~InclementWX,data=data.expanded, main="Profit based on
Weather",xlab="Inclement Weather (Defined as snow or rain)",ylab="Profit to Rivermen","Effect
of Inclement Weather on Rivermen Profit")
boxplot(RESPONSE~TicketSales,data=data.expanded, xlab="Total Ticket Sales Revenue",
ylab="Profit to Rivermen", main="Profit and Ticket Sales Revenue")
boxplot(RESPONSE~Drop.Total,data=data.expanded, xlab="Total Attendance", ylab="Profit to
Rivermen", main="Profit and Drop Total")
boxplot(RESPONSE~MeanTemp,data=data.expanded, xlab="Mean Temperature (degrees F)",
ylab="Profit to Rivermen", main="Profit and Temperature")
boxplot(TicketSales~Record, data=data.expanded, main="Ticket Sales and Overall Record",
xlab="Overall Record (Games Above / Below .500)", ylab="Ticket Sales")
```

```
boxplot(Concessions~Comp, data=data.expanded, main="Concession Revenue and Comped
Tickets", xlab="Free Tickets Redeemed", ylab="Concession Revenue")
boxplot(Concessions~Secondary, data=data.expanded, range=100,main="Concession
Revenue and Purchased Tickets", xlab="Paid Attendance", ylab="Concession Revenue")
boxplot(Concessions~Receipts, data=data.expanded ,main="Concession Revenue and
Receipts", xlab="Ticket Receipts", ylab="Concession Revenue")



indexOfResponse = which(colnames(data.expanded)=="RESPONSE")
x = as.matrix(data.expanded[,-indexOfResponse])
y = data.expanded[,indexOfResponse]

modelLasso = cv.glmnet(x,y)
plot(modelLasso, main="Overall Model")
print(coef(modelLasso))


#------------------
modelLM1 = lm(RESPONSE~Paid+Receipts+Concessions,data.expanded)
summary(modelLM1)


#------------------
# A model with more actionable varables

variables = c("OpponentColumbus",     "OpponentFayetteville", "OpponentHuntsville",
"icebears",
          "flyers" ,              "gators" ,              "OpponentKnoxville",    "OpponentLouisiana" ,
            "OpponentMacon"  ,      "OpponentMississippi" , "OpponentPensacola"  ,
"OpponentRiverkings" ,
            "DOWeekSaturday"   ,    "DOWeekSunday"   ,      "DOWeekThursday"   ,
"DOWeekTuesday" ,
          "Season2"    ,          "Season3"    ,          "MeanTemp",
          "InclementWX"        ,  "Record" ,              "DOWeekFriday"   , "Comp2Paid"    ,
"RESPONSE"  )

data2 = data.expanded[,variables]


indexOfResponse = which(colnames(data2)=="RESPONSE")
x = as.matrix(data2[,-indexOfResponse])
y = data2[,indexOfResponse]
```

```
modelLasso = cv.glmnet(x,y,nfolds=10)
plot(modelLasso)
print(coef(modelLasso))
mode

modelLasso2 = glmnet(x,y)
plot(modelLasso2,label=T)


modelLM2 = lm(RESPONSE~DOWeekSaturday+DOWeekSunday+Record+Comp2Paid,data2)
print(summary(modelLM2))


#--------------- Checking effect of paid and comp on concessions

variables2 = c("OpponentColumbus",     "OpponentFayetteville", "OpponentHuntsville",
"icebears",
            "flyers" ,              "gators" ,              "OpponentKnoxville",    "OpponentLouisiana" ,
            "OpponentMacon"  ,      "OpponentMississippi" , "OpponentPensacola"  ,
"OpponentRiverkings" ,
            "DOWeekSaturday"   ,    "DOWeekSunday"   ,      "DOWeekThursday"   ,
"DOWeekTuesday" ,
            "Season2"    ,          "Season3"    ,          "MeanTemp",
            "InclementWX"        ,  "Record" ,              "DOWeekFriday"   ,
"Paid","Concessions","Comp"  )

data3 = data.expanded[,variables2]


indexOfResponse = which(colnames(data3)=="Concessions")
x = as.matrix(data3[,-indexOfResponse])
y = data3[,indexOfResponse]

modelLasso = cv.glmnet(x,y,nfolds=10)
plot(modelLasso)
print(coef(modelLasso))

#---------------- Checking effect of paid and comp on concessions but using Comp2Paid instead
of Comp as a variable

variables2 = c("OpponentColumbus",     "OpponentFayetteville", "OpponentHuntsville",
"icebears",
```

```
            "flyers" ,              "gators" ,              "OpponentKnoxville",    "OpponentLouisiana" ,
            "OpponentMacon"  ,      "OpponentMississippi" , "OpponentPensacola"  ,
"OpponentRiverkings" ,
            "DOWeekSaturday"   ,    "DOWeekSunday"   ,      "DOWeekThursday"   ,
"DOWeekTuesday" ,
            "Season2"    ,          "Season3"    ,          "MeanTemp",
            "InclementWX"        ,  "Record" ,              "DOWeekFriday"   ,
"Paid","Concessions","Comp2Paid"  )

data3 = data.expanded[,variables2]


indexOfResponse = which(colnames(data3)=="Concessions")
x = as.matrix(data3[,-indexOfResponse])
y = data3[,indexOfResponse]

modelLasso = cv.glmnet(x,y,nfolds=10)
plot(modelLasso)
print(coef(modelLasso))

modelLasso2 = glmnet(x,y)
plot(modelLasso2,label=T)
modelLM3=lm(Concessions~flyers+OpponentKnoxville+OpponentMississippi+DOWeekSunday
+DOWeekThursday+Record+Paid+Comp2Paid, data3)
summary(modelLM3)
```